DATA MINING II
Version A
1st Term exam
June 18th , 2024

MGI

Student Nº:

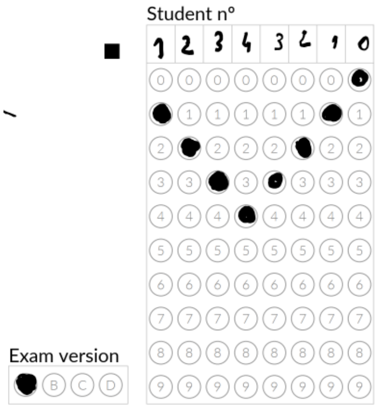
Name:

Instructions

1. This exam consists of 32 multiple-choice questions. Each question is worth 0.625 points, and wrong answers have a **penalty of** $\frac{1}{3} \times 0.625$.
2. The **answers** should be written on the **specific answer sheet** provided; both the answer sheet and the exam must be handed to the professor at the end.
3. In the **answer sheet,** please:
   a. Fill in your name, the class name (ML) and the date

Name	Course	Date

   b. Fill in the circles regarding the exam version and your student number. In the example below, student 12343210 answers version A of the exam.



   c. **Fill in** the correct circle for each question.
   d. If the answer is not clear, please ask for another answer sheet. We will assume incorrect answers that are **unclear or unreadable**.
   e. Each **question** has **one** possible **answer**. More than one option selected for a question results in an incorrect answer.
4. Any supporting material (books, notebooks, laptop, calculator and cell phone) and information exchange with colleagues **is prohibited**.
5. Please **also write your name and student number** in this exam header
6. The exam duration **is 50 minutes**.

1. In supervised learning, we have two different types of problems:
   a) Non-Parametric and semi-parametric problems
   b) Regression and Classification problems
   c) Predictive and clustering problems
   d) Overfitting and non-overfitting problems

2. The non-parametric models:
   a) Are constrained by a functional form
   b) Are usually harder to explain compared to parametric models
   c) Requires less data than parametric models
   d) Are usually associated with a poor fit, when compared to parametric models

3. Overfitting occurs when a model:
   a) Correctly predicts the training data but not the test data
   b) Correctly predicts the training data and the test data
   c) Correctly predicts the test data but not the validation data
   d) Correctly predicts the validation data but not the training data

4. In CRISP-DM, data preparation includes:
   a) Transform and clean the data
   b) Identify data quality problems such as outliers
   c) Define the objectives and requirements needed for the project
   d) Both a) and b) are correct

5. Concerning validation and test data sets:
   a) The test dataset is used to evaluate the performance during training
   b) Validation data set is applied eo evaluate overfitting and to fine-tune the hyperparameters of the model
   c) Validation and test data sets have the same purpose
   d) The test set should be used interchangeably with the validation set during the training process.

6. One of the disadvantages associated with the K-Fold cross-validation method is:
   a) It only provides a single estimate of the performance of the model
   b) We need always to apply 10 folds
   c) We are not able to apply stratification
   d) It is computationally expensive compared to the Hold-Out method

7. In a dataset with 1000 observations and 20 independent variables, if I apply K-Fold Cross-Validation with 10 folds:
   a) Each model created uses 900 observations for train, 100 observations for test, and we create 10 models
   b) Each model created uses 18 independent variables to train, 2 independent variables to test, and we create 1000 models
   c) Each model created uses 900 observations to train, 100 observations to test, and we create 20 models
   d) Each model created uses 18 independent variables to train, 2 independent variables to test, and we create 10 models.

8. In the example of predicting the drug dosage in milimeters to a disease:
   a) This is a regression problem and the drug dosage is the dependent variable
   b) This is a regression problem and the drug dosage is the independent variable
   c) This is a classification problem and the drug dosage is the dependent variable
   d) This is a classification problem and the drug dosage is the independent variable

9. What is a “pure” leaf in a classification decision tree?
   a) A leaf node with the highest number of observations
   b) A leaf node with the lowest number of observations
   c) A leaf node where all the observations belong to a single class
   d) None of the options is correct

10. Which of the following is a true statement about the K-Nearest Neighbors (KNN) algorithm?
    a) KNN is prone to overfitting if the value of K is too large.
    b) KNN can only be used for classification tasks and not for regression.
    c) The choice of the distance metric can significantly affect the performance of a KNN model.
    d) KNN requires a training phase where the model parameters are optimized using gradient descent.

11. The curse of dimensionality is related to:
    a) The high number of independent variables in a dataset that could lead to sparse data
    b) The high number of possible values in a categorical variable that could lead to biased results
    c) The low number of features that do not allow to create predictive models powerful enough
    d) All options are wrong

12. When analyzing a dataset containing questions about the age of the inquirees, we found a correlation between the missing values on age and the gender of the person. We are in a situation with:
    a) Missing values completely at random
    b) Missing values at random
    c) Missing values not at random
    d) Correlated Missing values

13. The MinMax normalization:
    a) Guarantees that all data points lie between 0 and 1
    b) Guarantees that all data points lie between a specified range
    c) Guarantees that the mean value of all data points is 0 and the standard deviation is 1
    d) Guarantees that the mean value of all data points is 1 and the standard deviation is 0

14. If I have a spearman correlation of -1 between variable A and variable B, that means that:
    a) Variable A and Variable B have a perfect negative linear relationship
    b) Variable A and Variable B have no relationship at all
    c) Variable A and Variable B have a negative relationship
    d) The higher the values of the variable A, the higher the values of Variable B

15. What is the primary purpose of post-pruning in decision trees?
    a) To select the best attribute for splitting the root node
    b) To remove noisy data points from the training set
    c) To reduce complexity and overfitting of the tree
    d) To make the algorithm of decision tree faster in the training stage

16. In feature selection, the filter methods:
    a) Use Machine Learning models to try to reduce the dimensionality of the dataset
    b) Use Machine Learning models and statistical approaches to try to reduce the dimensionality of the dataset
    c) Have the final purpose of identifying variables with high correlation and removing those
    d) Use statistical approaches to try to reduce the dimensionality of the dataset

17. In the OneR algorithm:
    a) Our decision is based only on the target.
    b) Our decision is based on the feature with the highest information gain.
    c) Our decision is based on the feature with the lowest entropy.
    d) Our decision is based only on one independent variable and the target.

18. In a regression tree:
    a) If MSE is used to choose the best split, the predicted value at each leaf corresponds to the mean value of the target of all the observations on that leaf
    b) If MSE is used to choose the best split, the predicted value at each leaf corresponds to the median value of the target of all the observations on that leaf
    c) If MAE is used to choose the best split, the predicted value at each leaf corresponds to the most frequent value of the target of all the observations on that leaf
    d) None of the options is correct

19. Why might you choose to use a KD Tree Algorithm for the nearest neighbor search over the Brute Force Algorithm?
    a) KD Tree Algorithm is faster in the prediction stage, in low to moderate dimensional spaces.
    b) KD Tree Algorithm can handle missing values.
    c) KD Tree Algorithm can work with categorical data.
    d) KD Tree Algorithm is more accurate.

20. In stacking, a base learner is?
    a) A model trained on the original data whose outputs are used as inputs to the meta-learner.
    b) A model trained on a bootstrap sample of the original data and whose outputs are used to vote on the ensemble's final prediction.
    c) a model that uses the out-of-sample predictions of other models and is assigned a weight by the meta-learner.
    d) A model that takes as input the outputs of the meta-learner and uses them to predict the ensemble's output.

21.Which of the following is an assumption of the Naive Bayes classifier?

a) Attributes are correlated
b) Attributes are conditionally independent
c) Complex relationships exist between attributes
d) There are no redundant attributes

22.What is the key difference between bagging and boosting ensemble methods?

a) Diversity of models
b) Number of models
c) Independence vs dependence of models
d) Hard vs easy data point focus

23.Which algorithm utilizes bootstrap sampling of data to create multiple prediction models?

a) AdaBoost
b) Gradient boosting
c) Stacking
d) Bagging

24.Which algorithm uses weak learner decision stumps as base models?

a) Random forest classifier
b) Logistic regression
c) Stacking
d) AdaBoost

25.Which of the following is a disadvantage of Leave-One-Out method?

a) The variance of the resulting estimate is smaller when the number of samples is higher
b) It is not appropriate for large datasets
c) It only provides a single estimate
d) The different test sets overlap in some of the observations

26.After creating a model, we obtain the following confusion matrix:

		Ground Truth	
		No	Yes
Predicted Value	No	10	35
	Yes	5	50

We can say that:

a) The accuracy of the model is 0.4
b) The number of False Positives is equal to 35
c) Accuracy is not a good measure to evaluate our model since the dataset is unbalanced
d) None of the options is correct

27. In Gaussian Naïve Bayes Classifier:
    a) We assume that the attributes are conditionally dependent on each other
    b) The presence of correlated features can lead to better results
    c) We are able to identify outliers and remove them
    d) We assume that the variables follow a normal distribution

28. We have a dataset with 4 independent variables – HoursStudy, PresenceClass, ProjectGrade, and HomeworksGrade. We want to predict if a student is going to pass the course (Pass = 1, Not pass = 0). The gini index in HoursStudy is 0.3, in PresenceClass is 0.2, in ProjectGrade is 0.4 and in HomeworkGrade is 0.8. The variable that we should use for the first split in our decision tree is:
    a) HoursStudy
    b) PresenceClass
    c) ProjectGrade
    d) HomeworkGrade

29. Which of the following events marked the "renaissance" of neural networks in the 1980s?
    a) The invention of the first artificial neuron
    b) The introduction of the "Perceptrons" book by Minsky
    c) The re-invention of the "Back-Propagation" algorithm
    d) The creation of the first self-driving car

30. What is the primary objective of the backpropagation algorithm?
    a) To randomly initialize the weights
    b) To minimize the error between the target and the output
    c) To calculate the total input to a neuron
    d) To reinforce synapses that achieve good results

31. What is the general idea behind the learning process in a neural network?
    a) Synapses that lead to unwanted results should be reinforced
    b) Synapses that help achieve good results should be weakened
    c) Synapses that help achieve good results should be reinforced
    d) Weights are adjusted randomly without any specific rule

32. Why are neural networks considered universal approximators?
    a) They can solve linear problems only
    b) They can approximate any function given sufficient neurons and layer
    c) They can only handle discrete-valued functions
    d) They are always more accurate than other machine learning models